

ENHANCING THE GERMAN TRANSMISSION GRID THROUGH DYNAMIC LINE RATING

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ABSTRACT

The deployment of new renewable energy sources far from load centers is leading to an increasing need for grid capacity. A notable example can be found in Germany, where most of the wind energy is generated in the North while there is significant industrial demand in the South [1]. Whereas the current federal government plans to add ~70 GW wind power capacity until 2030, only a few grid expansion projects have been realized [2, 3]. Moreover, most of these projects have been delayed due to administrative problems and protest activities [4]. This trend will likely exacerbate transmission system congestion, which may lead to more grid instability and curtailment. There is therefore a strong incentive to exploit existing grid infrastructure more fully.

Traditionally, transmission line capacity is calculated assuming unfavorable, static weather conditions such as 40°C ambient temperature and 0.6 m/s wind speed [5]. This is referred to as Static Line Rating (SLR). By design, SLR underestimates the capacity of a transmission line and, when implemented in practice, leads to an overestimated transmission capacity. In contrast, Dynamic Line Rating (DLR) calculates the line capacity taking into account the prevailing weather conditions. Cold weather and wind cool overheated transmission lines, enabling the thermal rating to be raised. This results in key benefits in cost-efficiency, congestion reduction and better wind power integration [6]. So far, the literature provides case studies considering DLR in small scale Energy System Optimization Models (ESOMs) [7–9]. However, it is lacking of an extensive evaluation and assessment of DLR being applied to a higher scale ESOM considering capacity expansion and a CO₂ target.

This paper addresses this gap. We model and analyze the impact of DLR on the optimal dispatch and deployment of power generators in the German electricity sector. The underlying model is created from the PyPSA-Eur framework [10], and for Germany consists of 40 representative nodes and considers a time horizon of one year with an hourly resolution. To model the effect of DLR on the power system, we use the IEEE standard [11] with highly resolved historical weather data implemented in the Atlite software [12]. The advantages of the DLR are illustrated by comparison with an SLR-based system. Therefore, we analyze and compare curtailment, grid congestion as well as the optimal generator deployment. Preliminary results show a cost reduction of 5-10 % when applying DLR instead of SLR while using the same transmission system. Furthermore, DLR leads to ~15 % higher wind power deployment and ~75 % higher average power flow compared to SLR.

REFERENCES

- [1] Olaf Kühne and Florian Weber. "Conflicts and negotiation processes in the course of power grid extension in Germany". In: *Landscape Research* 43.4 (May 19, 2018), pp. 529–541. ISSN: 0142-6397, 1469-9710. DOI: [10.1080/01426397.2017.1300639](https://doi.org/10.1080/01426397.2017.1300639). URL: <https://www.tandfonline.com/doi/full/10.1080/01426397.2017.1300639> (visited on 01/21/2022).
- [2] Federal Ministry of Economic Affairs and Climate Action. *Germany's current climate action status*. Jan. 13, 2022, p. 37. URL: <https://www.bmwi.de/Redaktion/EN/Downloads/E/germany-s-current-climate-action-status.html> (visited on 01/28/2022).
- [3] Prof Dr Bruno Burger. *Energy-Charts*. <https://energy-charts.info/?l=de&c=DE>. (Visited on 01/28/2022).
- [4] Mario Neukirch. "Protests against German electricity grid extension as a new social movement? A journey into the areas of conflict". In: *Energy, Sustainability and Society* 6.1 (Dec. 2016), p. 4. ISSN: 2192-0567. DOI: [10.1186/s13705-016-0069-9](https://doi.org/10.1186/s13705-016-0069-9). URL: <https://energysustainsoc.biomedcentral.com/articles/10.1186/s13705-016-0069-9> (visited on 01/21/2022).
- [5] E. Fernandez, I. Albizu, M.T. Bedialauneta, A.J. Mazon, and P.T. Leite. "Review of dynamic line rating systems for wind power integration". In: *Renewable and Sustainable Energy Reviews* 53 (Jan. 2016), pp. 80–92. ISSN: 1364-0321. DOI: [10.1016/j.rser.2015.07.149](https://doi.org/10.1016/j.rser.2015.07.149). URL: <https://linkinghub.elsevier.com/retrieve/pii/S1364032115007960> (visited on 01/19/2022).
- [6] F. Gülşen Erdiñç, Ozan Erdiñç, Recep Yumurtacı, and João P. S. Catalão. "A Comprehensive Overview of Dynamic Line Rating Combined with Other Flexibility Options from an Operational Point of View". In: *Energies* 13.24 (Dec. 12, 2020), p. 6563. ISSN: 1996-1073. DOI: [10.3390/en13246563](https://doi.org/10.3390/en13246563). URL: <https://www.mdpi.com/1996-1073/13/24/6563> (visited on 12/08/2021).
- [7] Yanlin Li, Bo Hu, Kaigui Xie, Lingfeng Wang, Yingmeng Xiang, Ruosong Xiao, and Dezhuang Kong. "Day-Ahead Scheduling of Power System Incorporating Network Topology Optimization and Dynamic Thermal Rating". In: *IEEE Access* 7 (2019). Conference Name: IEEE Access, pp. 35287–35301. ISSN: 2169-3536. DOI: [10.1109/ACCESS.2019.2904877](https://doi.org/10.1109/ACCESS.2019.2904877).
- [8] Morteza Dabbaghjamesh, Abdollah Kavousi-Fard, and Shahab Mehraeen. "Effective Scheduling of Reconfigurable Microgrids With Dynamic Thermal Line Rating". In: *IEEE Transactions on Industrial Electronics* 66.2 (Feb. 2019), pp. 1552–1564. ISSN: 0278-0046, 1557-9948. DOI: [10.1109/TIE.2018.2827978](https://doi.org/10.1109/TIE.2018.2827978). URL: <https://ieeexplore.ieee.org/document/8340159/> (visited on 01/20/2022).
- [9] Nicola Viafora, Kateryna Morozovska, Syed Hamza Hasan Kazmi, Tor Laneryd, Patrik Hilber, and Joachim Holbøll. "Day-ahead dispatch optimization with dynamic thermal rating of transformers and overhead lines". In: *Electric Power Systems Research* 171 (June 2019), pp. 194–208. ISSN: 0378-7796. DOI: [10.1016/j.epsr.2019.02.026](https://doi.org/10.1016/j.epsr.2019.02.026). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0378779619300902> (visited on 01/19/2022).
- [10] Jonas Hörsch, Fabian Hofmann, David Schlachtberger, and Tom Brown. "PyPSA-Eur: An Open Optimisation Model of the European Transmission System". In: *Energy Strategy Reviews* 22 (Nov. 2018), pp. 207–215. ISSN: 2211-467X. DOI: [10.1016/j.esr.2018.08.012](https://doi.org/10.1016/j.esr.2018.08.012). arXiv: [1806.01613](https://arxiv.org/abs/1806.01613). URL: <http://arxiv.org/abs/1806.01613> (visited on 01/21/2022).
- [11] *IEEE Standard for Calculating the Current-Temperature Relationship of Bare Overhead Conductors*. ISBN: 9780738188874. IEEE, 2012. DOI: [10.1109/IEEESTD.2013.6692858](https://doi.org/10.1109/IEEESTD.2013.6692858). URL: <https://ieeexplore.ieee.org/document/6692858> (visited on 01/21/2022).
- [12] Fabian Hofmann, Johannes Hampp, Fabian Neumann, Tom Brown, and Jonas Hörsch. "atlite: A Lightweight Python Package for Calculating Renewable Power Potentials and Time Series". In: *Journal of Open Source Software* 6.62 (June 24, 2021), p. 3294. ISSN: 2475-9066. DOI: [10.21105/joss.03294](https://doi.org/10.21105/joss.03294). URL: <https://joss.theoj.org/papers/10.21105/joss.03294> (visited on 01/21/2022).